JOHNSON PREM B

Mechanical Design Engineer

+91 96778 69242 | johnsonprem2002@gmail.com | Bengaluru, India | linkedin.com/in/johnson-prem-0616152ab

PROFESSIONAL SUMMARY

Mechanical Engineer Associate with experience in CAD design, EV battery concept development, product configuration, and CAD data management. Skilled in Creo, SolidWorks, AutoCAD, Fusion 360, Windchill, Options & Variants, reusable subassembly development, BOM management, and engineering documentation. Currently supporting EV battery concept work for Sankalp Innovation and configurable product structures for Fluence. Looking for a Mechanical Design or Manufacturing Engineer role where I can contribute to practical product development, CAD design, and configuration management.

TECHNICAL SKILLS

Mechanical Design:	3D CAD Modelling, Mechanical Assemblies, Technical Drawings, Sheet Metal Design, GD&T, Design Modification
EV & Energy Systems:	EV Battery Concept Design, Battery Packaging Concepts, Product Configuration
Engineering & Product Data:	BOM Management, CAD Data Management, CAD Restructuring, Product Configuration, Options & Variants, Reusable Subassemblies, Engineering Documentation
CAD & Engineering Software:	Creo, SolidWorks, AutoCAD, Fusion 360, Ansys Workbench, Windchill, Microsoft Office
Manufacturing & Quality:	Manufacturing and Fabrication Fundamentals, Assembly Concepts, Quality Inspection, Inspection Documentation

PROFESSIONAL EXPERIENCE

CADOpt Technologies | February 2026 – Present

Design Engineer Associate

Project: Fluence BESS India Pvt. Ltd | Product Configuration & Options and Variants

Key Contributions

- Worked on Options & Variants activities for Fluence BESS products.
- Created reusable subassemblies to support different product variants and improve CAD reuse.
- Worked with options, choices, rules, and variant specifications within Windchill.
- Supported configurable product structures and reusable CAD subassembly development.
- Structured CAD data to improve reuse across different product variants.
- Supported BOM and product-structure activities for configurable BESS products.
- Reviewed CAD structures and identified issues related to variant configuration.
- Collaborated with engineering teams to support configuration requirements and design changes.
- Maintained engineering and CAD data associated with configurable product structures.
- Applied product configuration concepts to create reusable design solutions.

Impact

- Developed reusable subassembly concepts to support multiple BESS product variants.
- Supported configurable product structures through Windchill Options & Variants.
- Contributed to structured CAD reuse and product configuration for BESS applications.

Project: Sankalp Innovation | EV Battery Concept Design

Key Contributions

- Supported electrical battery concept design activities for electric vehicle applications.
- Worked on CAD and engineering activities related to EV battery concepts and design requirements.
- Supported development and modification of battery-related design concepts based on project requirements.
- Worked with engineering data and CAD information during concept development activities.
- Collaborated with engineering teams to support design requirements and project deliverables.
- Applied mechanical engineering principles while supporting EV battery concept development.

Impact

- Contributed to EV battery concept development by connecting project requirements with CAD-based engineering work.

KWALITY Coil Products | July 2024 – August 2024

Intern

Oil-cell Quality Testing

- Studied oil-cell quality inspection procedures and applicable standards.
- Participated in quality control activities and professional inspection sessions.
- Supported quality-related documentation and reporting.
- Developed practical understanding of engineering quality and inspection procedures.
- Observed inspection practices and quality requirements applicable to engineering products.

EDUCATION

B.E. Mechanical Engineering

Latha Mathavan Engineering College, Madurai | Anna University | 2025

Diploma in Mechanical Engineering

St. Joseph's Polytechnic College | 2021

ACADEMIC PROJECTS

Design and Fabrication of Inline Pipe Inspection Robot | B.E. Mechanical Engineering Project

- Designed an inline pipe inspection robot using SolidWorks for 3D modelling and mechanical development.
- Developed mechanical components and assembly concepts for the inspection robot.
- Integrated a camera-based system for real-time visual inspection inside pipelines.
- Designed the system to identify potential cracks, corrosion, and blockages inside pipelines.
- Worked on integrating mechanical and inspection-system requirements during project development.

Frictionless Braking System | Diploma Project

- Developed a braking system concept focused on reducing mechanical friction.
- Studied design considerations related to safety, reliability, and component longevity.
- Applied mechanical engineering principles during concept development.

Certifications

- Participated in 4 Hours Intensive Live workshop on Mechanical Design organized by VCADD Engineering Solution.
- Completed Product Design Engineering and Modelling Course by AUTODESK.
- One day workshop on Industrial Grade 3D Printing.
- Certifications in CAD Software training for Creo, AutoCAD, SolidWorks, Ansys, Fusion360.

LANGUAGES - English, Tamil